

Possible Co-Evolution of Polyglutamine and Polyproline in Huntingtin Protein: Proline-Rich Domain as Transient Folding Chaperone

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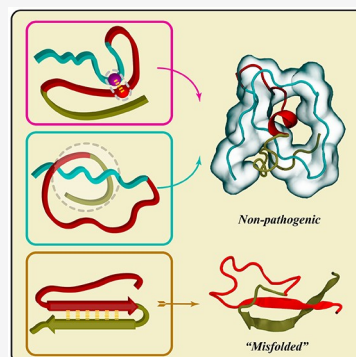


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ABSTRACT: Huntington's disease is an inherited neurodegenerative disorder caused by the overduplication of CAG repeats in the Huntingtin gene. Recent findings revealed that among the orthologs, the expansion of CAG repeats (polyQ) in the Huntingtin gene occurs in tandem with the duplication of CCG repeats (polyP). However, the molecular mechanism of this possible co-evolution remains unknown. We examined the structures of Huntingtin exon 1 (HttEx1) from six species along with five designed mutants. We found that the polyP segments "chaperone" the rest of the HttEx1 by forming ad hoc polyP binding grooves. Such a process elongates the otherwise poorly solvated polyQ domain, while modulating its secondary structure propensity from β -strands to α -helices. This chaperoning effect is achieved mostly through transient hydrogen bond interactions between polyP and the rest of HttEx1, resulting in a striking golden ratio of $\sim 2:1$ between the chain lengths of polyQ and polyP.



Huntington's disease (HD) is a neurodegenerative disease caused by an expansion of CAG codons in the Huntingtin (Htt) gene.^{1,2} The Htt gene-encoded Htt protein exists in all human cells and has been demonstrated to play vital roles in both the nervous system³ and embryonic development.⁴ Nonpathological Htt protein contains 6–35 glutamine residues (polyglutamines or polyQs) in its amino acid sequence. The number of glutamine-encoding CAG repeats in the Htt gene corresponds to the length of polyQ (or Q length) in the Htt protein and, when excessive, was found to be directly correlated with both the average onset age of HD symptoms^{5,6} and the tendency of mutant Htt proteins to aggregate.^{7,8} Despite controversial evidence of the pathology of HD,^{9,10} previous studies suggest that amorphous protein aggregates are more toxic than fibrils,¹¹ which associates the neurotoxicity of Htt protein with its aggregation rate. Furthermore, while the entire Htt protein consists of ~ 3142 residues,¹² the N-terminus of the protein, i.e., exon 1 (Ex1)-encoded HttEx1 protein, is commonly recognized as the pathogenic entity (with its Q length exceeding 36 residues) and has been targeted for clinical intervention.¹³

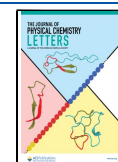
The mechanism behind the neurotoxicity of HttEx1 protein can be investigated via the properties of the protein's three distinct domains. The HttEx1 protein is an intrinsically disordered protein^{14,15} and is composed of an N-terminal 17-residue domain (N17), a polyQ domain, and a proline-rich domain (PRD) at the C-terminus. An isolated N17 is largely unstructured in solution but folds into α -helical conformations upon binding to membranes.^{15–17} The N17 domain has therefore been proposed to function as an anchor of the Htt

proteins to cell membranes.^{18–20} On the contrary, accumulating evidence suggests that when the polyQ domain serves as the central immobilized part, the N17 domain strongly promotes the aggregation of HttEx1.^{15,21} Specifically, the strong hydrogen bonding capability of glutamines leads to non-negligible intra- and intermolecular interactions, which result in the polyQ segments forming filaments (especially when the length of the glutamine segment is greater than seven residues²). Additionally, the HttEx1 aggregates are found to consist mainly of stacked β -sheets.²² The third domain of HttEx1 protein, the PRD, is mainly composed of several short segments of polyprolines (polyPs), connected by short hinge residue sequences containing glutamines (Q), leucines (L), and alanines (A). This domain has been identified as the major binding partner in HttEx1's extensive protein–protein interactions.^{23–29} Interestingly, it has been shown that the PRD holds the potential to alleviate the neurotoxicity of HttEx1 by weakening the protein's tendency to aggregate.^{11,30} Moreover, a previous study showed that replacing the PRD with a P₁₀KK segment has a negligible impact on the aggregation tendency of HttEx1,¹⁵ suggesting that a simplified sequence is also capable of fulfilling the protective function of a

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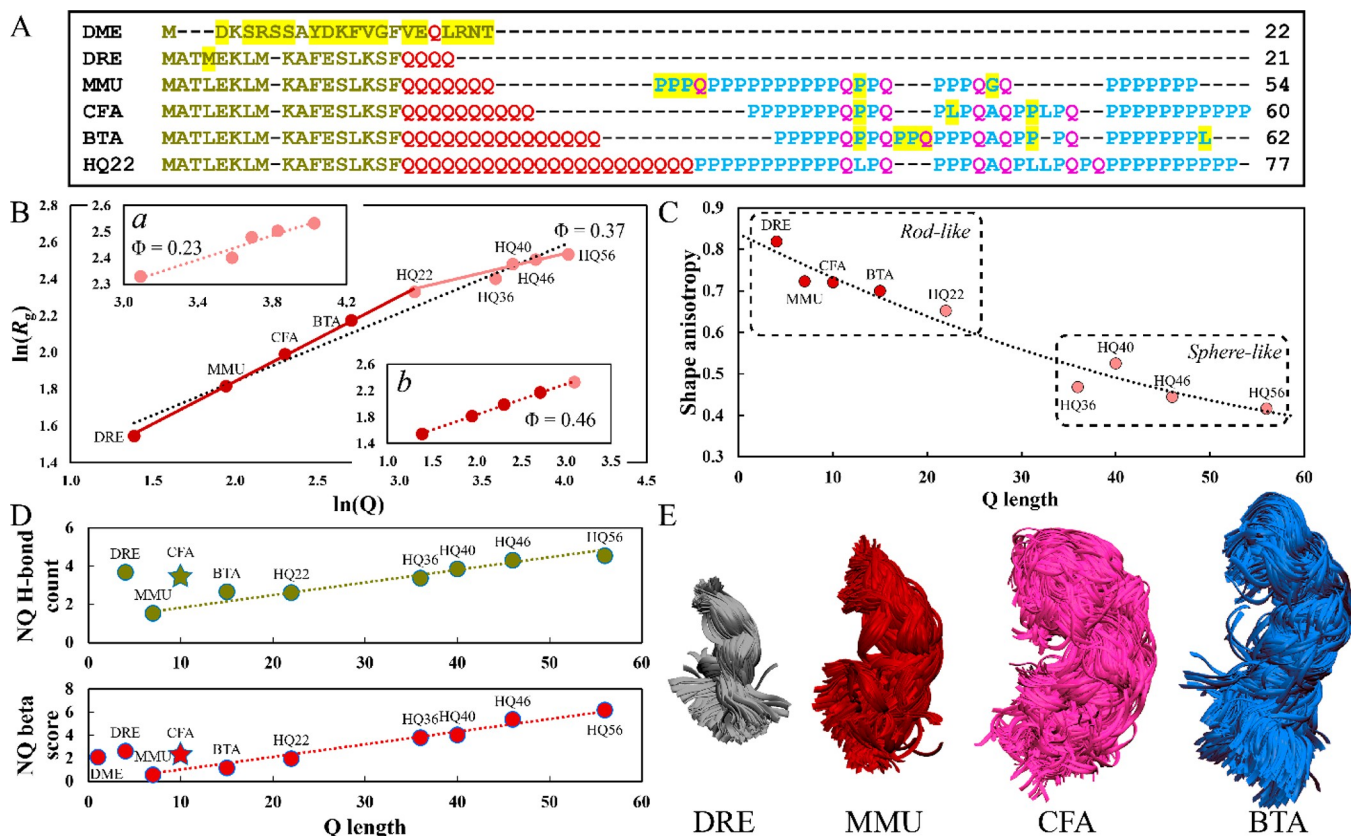


Figure 1. (A) Alignment of the sequences of the five additional forms of Htt exon 1 (DME, DRE, MMU, CFA, and BTA) with human Htt exon 1 with a Q length of 22 (HQ22). Note that forms of human Htt exon 1 with other Q lengths are also included in most of the analyses in this study (HQ36, HQ40, HQ46, and HQ56, taken from ref 46). The sequences are colored as follows: tan for the N-terminus (17 residues before polyQ except DME, therefore N17), red for polyQ, purple for Q residues in the C-terminus (proline-rich domain, therefore PRD), and cyan for the rest of the PRD. Residues different from the HQ22 sequence are highlighted in yellow. (B) Radii of gyration of polyglutamine (polyQ) vs the natural logarithm of the number of glutamines in polyQ (Q length) for different species. A linear fit is shown as a dotted black line with a slope Φ of 0.37. The red line fits all wild types with a Q length of greater than one (DRE, MMU, CFA, BTA, and HQ22), which is shown in the inset at the bottom right. The pink line fits the human polyQs (taken from ref 46), which is shown in the inset at the top left. (C) Shape anisotropy κ of the polyQs plotted vs Q length. The dotted line is a guide to the eyes. (D) NQ β score (a measurement of the likelihood of the combined N17-polyQ domain to form β -sheets; see [Methods in the Supporting Information](#) for a detailed definition; bottom) and hydrogen bond counts between N17 and polyQ (NQ H-bond counts, top) plotted vs Q length. NQ stands for the segment of N17 and polyQ of the simulated proteins. Dotted lines do not imply a linear relationship but rather serve as references to single out the “outliers”: DRE and CFA. (E) Ensemble of polyQ structures from four species. One hundred structures were randomly picked to generate these images. Structures of human polyQs can be found in ref 46.

native PRD. Other studies have suggested that the PRD provides connection-specific protection to HttEx1 only when PRD is located at the C-terminus of polyQ.³¹ It was later suggested that the PRD might provide *cis* protection to HttEx1 by assimilating polyQ with its signature polyproline II (PPII) structures through backbone-mediated torsional propagations.^{32–34} However, a separated polyQ and polyP model ($K_2Q_{40}X_3P_{10}K_2$, where X was alanine, valine, or glycine) retained a similar level of the inhibitory effect from polyP, while backbone torsional propagation was mitigated.³¹ Furthermore, Duennwald et al. showed that the toxicity of Q103-HttEx1 was completely eliminated when Q25-HttEx1 was constitutively expressed at a stronger level but such protection was nonexistent without the PRD, suggesting that PRD could antagonize polyQ toxicity in *trans*.³⁵ With regard to the protective length of polyP, it was suggested that polyP with six prolines (P_6) or longer can suppress the aggregation of polyQ, and P_6 has shown no difference in slowing the aggregation of Q_{40} from P_{10} .³¹ However, it has also been found that populations of western European descendants have a higher prevalence of HD and also a higher incidence of the

(CCG)₇ allele instead of the (CCG)₁₀ allele compared to Japanese, Chinese, and African Blacks.³⁶ A few computational studies were conducted to examine the protective role of PRD. For example, it was suggested that PRD reduced the monomeric β -content, therefore reducing the toxicity of polyQ, but no structural explanation was provided.³⁷ In a hypothesized aggregated form, PRD was also found to weaken the tendency to aggregate based on molecular dynamics (MD) simulations.³⁸ However, no structural details of this inhibitory mechanism were given in this study, either. In summary, there are seemingly contradictory experimental findings in the literature. Nevertheless, without a detailed structural study of the interplay between PRD and the rest of the subdomains of HttEx1, the role of the PRD remains puzzling.³⁹

We first investigate the protective role of the PRD with respect to the length of this domain and the other two domains in HttEx1. A gene sequencing analysis of HttEx1 indicates that in vertebrates, the sequence of the N17 domain is highly conserved while that of the polyQ domain displays variations in length.⁴⁰ Notably, the PRD is exclusive of the mammalian vertebrates where the polyQ domain is longer than that of the

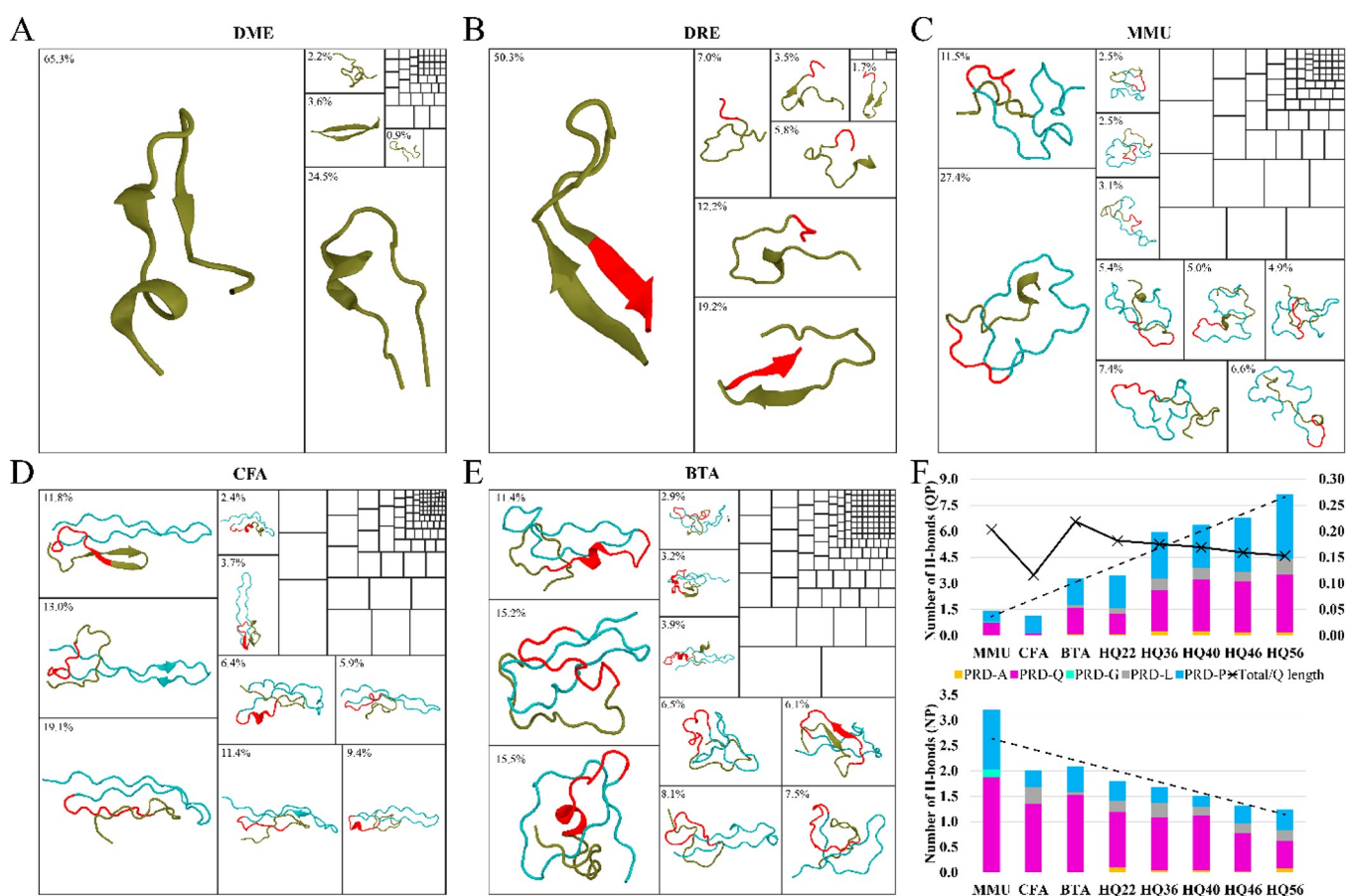


Figure 2. Tree maps depicting populous structures of five species: (A) DME, (B) DRE, (C) MMU, (D) CFA, and (E) BTA. The represented structures are cluster centers calculated using a root-mean-square deviation cutoff of 2 Å. The average H-bond count per time frame between polyQ and PRD is shown in the top panel of panel F, while the bottom panel shows that between N17 and the PRD. Contributions from each residue variety in the PRD are highlighted in purple for glutamines in the PRD (PRD-Q), cyan for prolines in the PRD (PRD-P), orange for alanines in the PRD (PRD-A), lime for glycines in the PRD (PRD-G), and silver for leucines in the PRD (PRD-L). The coloring is the same in Figures 3–5. Dashed lines are guides to the eyes.

nonmammalian counterparts.⁴⁰ Figure 1 and Figure S1 illustrate the sequence of HttEx1 orthologs for *Drosophila melanogaster* (DME), *Danio rerio* (DRE), *Monodelphis domestica* (MDO), *Mus musculus* (MMU), *Canis familiaris* (CFA), *Pan troglodytes* (PTR), *Loxodonta africana* (LAF), *Bos taurus* (BTA), *Sus scrofa* (SSC), and *Homo sapiens* (HQ22). Interestingly, even though the PRDs are vastly different among the mammalian vertebrate species, the general building blocks forming the PRD subdomains are the same: two long stretches of polyPs (4–13 prolines) connected by relatively flexible segments with short polyPs (1–3 prolines), glutamines, leucines, alanines, and threonines (Thr). Remarkably, the length of the primary polyP segment (defined as the first polyP stretch) is weakly correlated with the length of the polyQ domain (Figure S1D), which points toward the possibility of co-evolution between the two contiguous domains.⁴¹ More specifically, this correlation in length suggests that evolution might have chosen a co-duplication pattern between the glutamine-encoding CAG repeats and proline-encoding CCG repeats at the genetic level. Similar “co-duplication” events have been found in peptides that cause other trinucleotide repeat disorders (Table S1; also pointed out by Urbanek et al.⁴²). As the length of the polyQ domain is negatively correlated with the onset age of HD (that is, the longer the polyQ domain, the earlier the onset age), we speculate that the

PRD might have co-evolved with polyQ to rein in the potential toxicity of polyQ in more complex species. One hypothesis could be that if such co-evolution were directly linked between these two domains, polyQ and primary polyP might share a certain level of contact.^{36,41} However, it is unclear whether this correlation is coincidental and how the PRD could prevent and delay HD development. Interestingly, the PRD stretch in Huntingtin-associated protein 40 (HAP40-PRD) was found to similarly co-evolve with polyQ.⁴³ However, HAP40-PRD was found to be dispensable for the interactions between HAP40 and the full-length Htt. Although it is possible that these PRDs have co-evolved with other binding partners of Htt, herein we provide evidence at a molecular level that the interplay of the PRD and the other domains in HttEx1 might also be a feasible explanation for the positive correlation between the length of Htt-polyP and Htt-polyQ.

We then focus on the interdomain interactions between different domains in HttEx1 instead of investigating individual domains on their own. Due to the unstructured nature of monomeric HttEx1,^{14,15} understanding the interplay of its subdomains is intrinsically difficult. In this work, we investigate the interdomain interactions by considering the entire ensemble of HttEx1 structures from each species. This ensemble-based analysis is accomplished by applying replica exchange molecular dynamics (REMD) simulations in atom-

istic detail in explicit solvent (see [Methods in the Supporting Information](#) for details). Additionally, we designed multiple artificial proteins to reveal the exact functions of the polyP tracts and the glutamine residues in the PRD (glutamines-in-PRD). In our previous work, we demonstrated that the N17 domain may exacerbate the tendency of the polyQ domain.¹⁷ In this work, on the contrary, we observed that the PRD forms a “polyP cage” that “chaperones” the polyQ domain, which reduces polyQ’s overall tendency to form β -sheet structures. Furthermore, we identified that the length ratio between polyP and polyQ is also important in preventing the formation of β -sheet structures, where a golden ratio of polyQ length to polyP length may have been selected on an evolutionary basis and is optimal when this ratio stays at 2:1.

Long polyQs have an unexpected tendency to be sequestered from the aqueous environment (hydrophobicity) as observed via both fluorescence correlation spectroscopy⁴⁴ and atomistic MD simulations^{45,46} due to their strong capability to form intramolecular interactions, either side chain–side chain or side chain–backbone hydrogen bonds, despite being hydrophilic residues. Indeed, for monomeric polyQ in aqueous solutions, it was shown that the radius of gyration (an indicator of protein structure compactness) displays a Φ value of ~ 0.33 in its power law of $R_g = R_0 N^\Phi$ (where R_g is the radius of gyration of the polymer, N is the number of monomers, and R_0 and Φ are coefficients), implying that polyQ prefers the collapsed structure to minimize interactions with the surrounding solvent.⁴⁵ Previous research suggests that the transition temperature (T_θ) where free-standing polyQ acts like an ideal chain with a corresponding Φ value of 0.5 is ~ 410 K in an aqueous environment.⁴⁵ However, when we estimated the Φ value for polyQ domains alone from the five species used in this study ([Figure 1B](#), inset b), it was determined to be 0.46 at 310 K, indicating that the polyQ domain within the HttEx1 behaves like an “ideal chain” under this condition. This seeming disagreement in the transition temperature may be attributed to the presence of the flanking domains connected to polyQ (i.e., N17 and the PRD), which emphasizes the importance of including all three domains in HttEx1 when investigating the structural consequences of HD. From our analysis of the polyQ domain, we see that in the case of shorter Q lengths, interdomain interactions influence the polyQ segment to explore solvated “rod-like” conformations ([Figure 1E](#)). Notably, as the Q length increases, a transition in the conformational space sampled by the polyQ domain occurs, as reflected by the decline in the shape anisotropy values corresponding to the increment of the Q length ([Figure 1C](#)). This effect is particularly evident when the Q length is greater than 36 residues, that is, when the size of the PRD (38 residues) starts to become shorter than that of the polyQ domain. Consequently, the interdomain interactions may fail to elongate the polyQ domain, resulting in a change in its conformational behavior to now explore more sphere-like conformations ([Figure 1C](#)). This observation agrees with our previous study in which “super-compact” structural arrangements of polyQ were found in human HttEx1 with a Φ of 0.23.⁴⁶ The effects of the N17–polyQ interactions on polyQ elongation are supported by the “rod-like” conformations from the DRE ([Figure 1C](#)) and CFA Δ ct [CFA HttEx1 without the PRD ([Figure S2A](#))] systems, where the PRD is absent. However, in the case of greater Q lengths, the N17–polyQ interactions fail to elongate the polyQ domain, as indicated by the “sphere-like” structures adopted by the HQ22 Δ ct system

[human HttEx1 with 22mer polyglutamines without the PRD ([Figure S2B](#))]. However, restoring HQ22 Δ ct with a simplified version of the PRD, namely P₁₀A, does improve the shape anisotropy value ([Figure S2B](#)), which means the presence of the PRD is non-negligible. Therefore, the presence of both N17 and the PRD is essential to mediate the interactions with polyQ so as to favor the “rod-like” conformations for this domain.

Several studies, including circular dichroism (CD),¹⁵ NMR spectroscopy,¹⁴ and computational simulations ([Figure 2A–E](#) and [ref 46](#)), have verified the largely unstructured nature of the monomeric HttEx1 protein. X-ray scattering experiments²² revealed that the polyQ domain showed a strong propensity to adopt the extended β -strand conformation in its aggregated form. A study by Wagner et al.⁴⁷ suggested that β -sheet-rich seeds could induce the monomeric polyQ to misfold into an “aggregation-competent state” with β -sheet structural content. Kar et al. demonstrated that the artificially generated β -hairpin motif mediated polyglutamine amyloid formation.⁴⁸ Darnell et al.³³ proposed that polyQ in HttEx1 existed in a “tug of war” between two states: a β -sheet-rich state and a more disordered PPII-like state. Moreover, Nagai et al. used a pure glutamine chain to show that a transition of polyQ into a β -sheet conformation preceded amyloid fibril formation.⁴⁹ Comprehensively, a plausible hypothesis could be conjured from these studies that monomeric polyQ might transform from a nontoxic state to a toxic β -sheet-rich state. Such a mechanistic model has been generalized previously as the “N* theory”.^{50,51} However, in previous studies, flanking domains (N17 and PRD) were often ignored. Researchers have recognized that N17 aggravates the aggregation of this segment while the PRD alleviates this process.^{11,15,30} Meanwhile, N17 folded into an amphipathic α -helix on membranes,¹⁶ which in turn accelerated the aggregation of HttEx1.⁵² It was compelling to conclude that N17 accelerated HttEx1 through its formation of aggregation-prone α -helices.^{15,21,52} However, from the evidence that non-negligible β -sheet contents were found in N17 (in a crystal structure,⁵³ monomeric HttEx1,⁵⁴ and aggregated HttEx1⁵⁵), we hypothesize that there could be a second mechanism in terms of how N17 could aggravate HttEx1 aggregation. If we assume that the tendency to form β -strand structures by the polyQ segment positively correlates with the aggregation tendency of HttEx1, the N17 domain should increase the population of β -strands in this domain while the PRD should have the opposite effect. To test this assumption, we analyze a possible relationship between the interdomain interactions and the secondary structure content of the proteins. Accordingly, we find a clear positive correlation between the number of hydrogen bonds formed between N17 and polyQ and the N17–polyQ β score,⁵⁶ which is a quantitative measure of the N17–polyQ β -strand content [see [Methods](#) for the definitions (plotted in [Figure 1D](#))]. In short, the β score captures the resemblance of a sliding window of 3 + 3 residues to either a parallel β -structure or an antiparallel β -structure. For example, a hypothetical perfect β -hairpin with 112 residues (HQ57) would have a β score of $(112 - 2)/2 - 3 + 1 = 53$. Using the N17–polyQ β score and the number of N17–polyQ interactions as order parameters, we plotted the free energy landscape of HttEx1 proteins from all species examined in this study in [Figure S18](#) (with the number of hydrogen bonds between N17 and polyQ, i.e., NQ H-bonds) and [Figure S19](#) (with the number of heavy atom contacts between N17 and polyQ within 4.5 Å, i.e., NQ

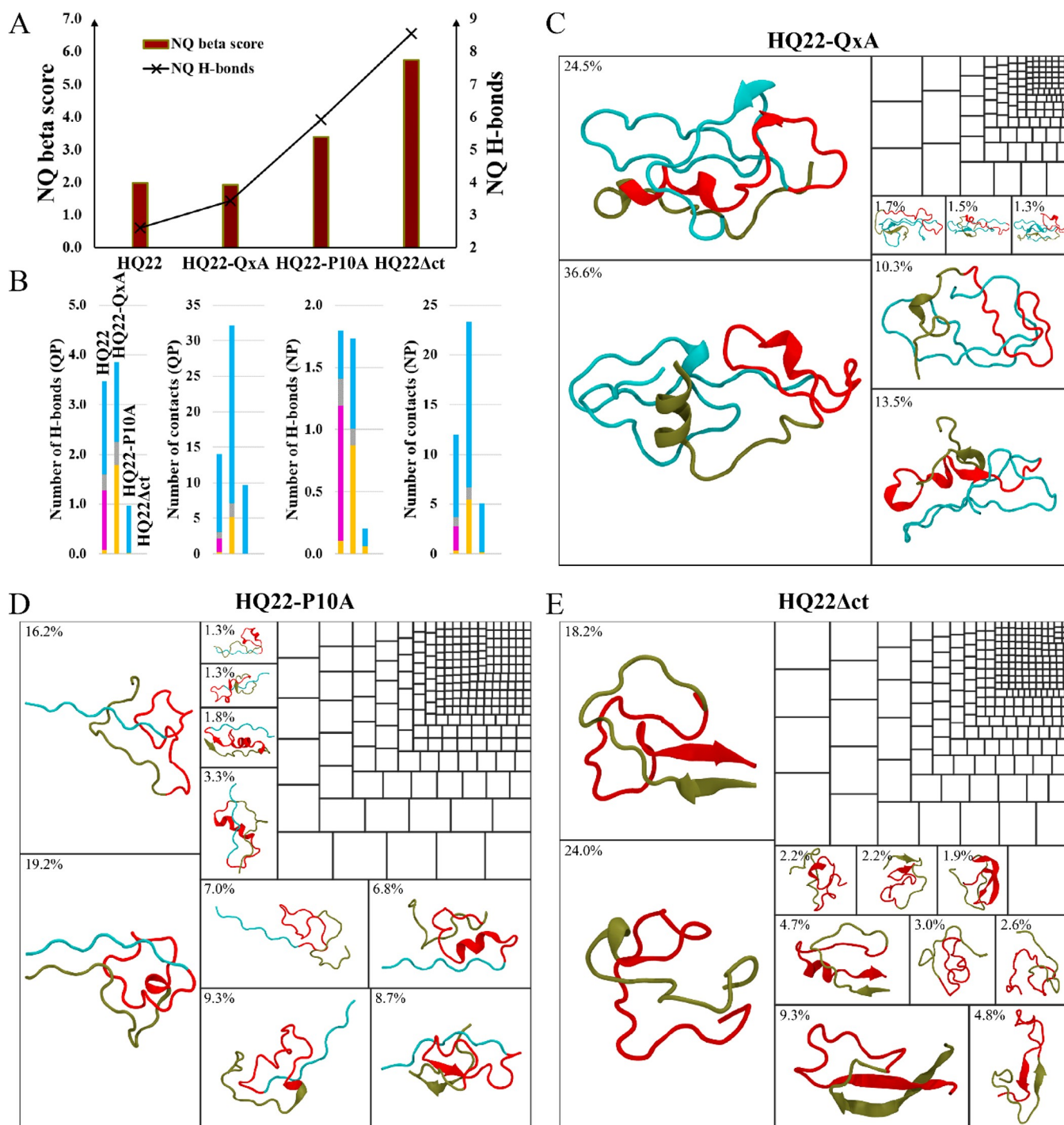


Figure 3. How PRD affects the folding states: a case study of HQ22. (A) NQ β scores and NQ H-bond counts (see the definitions in Figure 1) of HQ22, HQ22 with glutamines in the PRD (PRD-Qs) replaced with alanines (HQ22-QxA), HQ22 with the PRD replaced by P₁₀A (HQ22-P10A), and HQ22 with the PRD truncated (HQ22Δct). (B) Total numbers of QP H-bonds, QP contacts, NP H-bonds, and NP contacts for the four peptides (from left to right in each bar plot, HQ22, HQ22-QxA, HQ22-P10A, and HQ22Δct, respectively). (C–E) Tree maps (each box of which represents the population of a cluster) with cluster centers for HQ22-QxA, HQ22-P10A, and HQ22Δct, respectively.

contacts). A positive correlation is observed within all species, with an average covariance, R , of 0.40 between the N17–polyQ β score and the number of NQ H-bonds and 0.52 between the N17–polyQ β score and the number of NQ contacts. These results are a clear indication that the interdomain N17–polyQ interactions likely promote polyQ conformations with a high β -strand secondary structure content. As anticipated, deletion of the N17 domain from

HQ40 decreases the β -content of polyQ (unpublished data, consistent with previous studies^{37,57}). Moreover, the DRE and CFA systems consistently stand out as outliers above the trend lines corresponding to the N17–polyQ interactions, the N17–polyQ β score (Figure 1D), and the polyQ β residue count (Figure S3). These outliers are related to the functions of the PRD, which will be discussed below. With this evidence, it is

compelling to suggest that the N17–polyQ interactions increase the overall β -structure content of HttEx1.

Next, we evaluate the possible implication of the PRD in affecting the conformational space sampled by the polyQ domain. By investigating the interactions between the PRD and the other two domains, we demonstrate that the PRD plays an essential role in folding the polyQ domain. Furthermore, we provide mechanistic insights into the interplay of these domains and propose a co-evolution model in which the PRD might be carefully tailored by evolution to counteract the potential toxicity of long polyQ domains.

We start by examining the number of interdomain interactions between N17 and the PRD. We find that the number of H-bonds (Figure 2F) and the number of heavy atom contacts (Figure S4A) are both smaller in the presence of a longer polyQ domain. One may deduce a negative correlation between the number of N17–PRD interactions and the N17–polyQ β score. However, such a correlation could be a byproduct of the correlation between the Q length and N17–polyQ β score shown in Figure 1D. With greater Q lengths, a decrease in the number of N17–PRD interactions is expected due to the longer distance between the two domains. To remove the factor of Q length, we plotted the free energy profiles using N17–PRD interactions and the N17–polyQ β score as order parameters for all of the species examined in this study (Figure S20 using H-bonds and Figure S21 using contacts). Weak negative correlations are displayed within all species, with an average covariance R value of -0.24 between the number of N17–PRD H-bonds and the N17–polyQ β score and -0.32 between the number of N17–PRD contacts and the N17–polyQ β score.

Next, we examine the interdomain interactions between polyQ and the PRD. We found that the numbers of polyQ–PRD H-bonds (Figure 2F) in the DRE (where no PRD is present, so the value is 0) and CFA structures are below the trend lines. This is in contrast to the observation that the numbers of N17–polyQ H-bonds and the N17–polyQ β score (Figure 1D) of DRE and CFA are above the trend lines. A somewhat similar trend for the heavy atom contacts between polyQ and the PRD is also found (Figure S4B), with CFA and DRE as “outliers” below the trend line. The interesting fact that the DRE and CFA systems pose as outliers in all plots seems to confirm the critical role of the PRD regarding the possible reduction of the N17–polyQ β -content by the particular interactions between polyQ and the PRD. To further examine the impact of the polyQ–PRD interactions on the N17–polyQ β -content, we plot the free energy profiles using the polyQ–PRD interactions and the N17–polyQ β score as order parameters for all species examined in this study (Figure S22 using H-bonds and Figure 23 using contacts). Again, weak negative correlations seem to be displayed within all species, with an average covariance R value of -0.19 between the number of polyQ–PRD H-bonds and the N17–polyQ β score and -0.23 between the number of polyQ–PRD H-bonds and the N17–polyQ β score.

Finally, if we average the tallied number of interactions between the PRD and the rest of HttEx1 using all species, non-negligible amounts of polyQ–PRD interactions and N17–polyQ interactions are detected (1–7 H-bonds and 7–33 heavy atom contacts). This finding agrees with the results of previous smFRET experiments in which long-range interactions were found in HttEx1⁵⁸ and PRD in the HttEx1 fibril was shown to be immobilized and involved in intimate

interactions.²¹ These results differ from those of prior studies in which the PRD was conceived as a rigid rod in a “tadpole” that hardly interacted with other domains.^{54,58} More importantly, we observe that the glutamine residues in the PRD disproportionately contribute to the interdomain H-bond interactions (Figure 2F and Figure S4C,D), whereas the atomic contacts are dominated by the proline residues in the PRD (Figure S4A,B). In fact, the atomic contacts normalized by the number of glutamines in the PRD and prolines in the PRD indicate that their per-residue contributions are similar (Figure S4E,F), which leads to the following speculation. (i) Glutamines in the PRD may act as H-bond anchors that influence the polyQ conformations, and (ii) prolines in the PRD may act as “folding chaperones” by providing (relatively) rigid and hydrophobic “cages” for the N17 and polyQ domains as depicted in panels C and E of Figure 2. To verify this speculation, we designed the following systems on the basis of native HQ22: (1) HQ22-QxA, in which we replaced all glutamines with alanines (Ala) in the PRD of the HQ22 system to verify the function of these residues, (2) HQ22-P10A, in which we replaced the PRD of HQ22 with the simplified analogue P10A (P₁₀A) to test the necessity of polyP cages, and (3) HQ22 Δ ct, in which we deleted the entire PRD in HQ22 as a control.

First, when we mutate all glutamines to alanines in the PRD, the β -strand structural content of the polyQ and PRD segments increases (from 5.80 ± 0.04 in HQ22 to 6.46 ± 0.05 in HQ22-QxA), while the N17–polyQ β score remains roughly unchanged (Figure 3). This is because the glutamine side chains prevent the formation of backbone–backbone H-bonds, which are necessary for the appearance of high- β -content structures (only 16% of H-bonds between polyQ and glutamines in the PRD are backbone–backbone H-bonds), while alanine residues are restricted to establishing this type of polar interaction only via backbone atoms (58% of H-bonds between polyQ and alanines in the PRD are backbone–backbone H-bonds). These results are somehow surprising, because normally alanines are among the residues with the lowest β -sheet propensity, lower than that of glutamines.^{59,60} However, the effect of glutamines in the PRD surpasses that of alanines because of the capability of the glutamines to establish side chain-mediated H-bonds. Therefore, it is likely that evolution has reached some solution in selecting glutamines as the hinge residues in the PRD.

Second, we find that the simplified P₁₀A segment also induces the N17–polyQ complex to form an ad hoc polyP binding groove by providing both H-bond acceptors and hydrophobic interactions (Figure 3D). Interactions between P₁₀A and the N17–polyQ complex result not only in weaker overall N17–polyQ interactions but also in a weaker tendency of the N17–polyQ complex to form β -structures in HQ22-P10A than in the HQ22 Δ ct system (Figure 3A,B). Therefore, the chaperoning effect may be partially provided by the simplified PRD analogue (P₁₀A) even though it does not measure up to that of the full PRD.

To further decipher the interacting pattern of the PRD, we calculate the average number of H-bonds and average number of residue contacts for each residue in terms of the N17–PRD and polyQ–PRD interdomain interactions. We see that the N17–PRD interactions are divided between hydrophilic interactions from glutamate (E), serines (S), and threonines (T) and hydrophobic interactions from alanines (A), leucines (L), and phenylalanines (F) (Figures S5 and S6). Meanwhile,

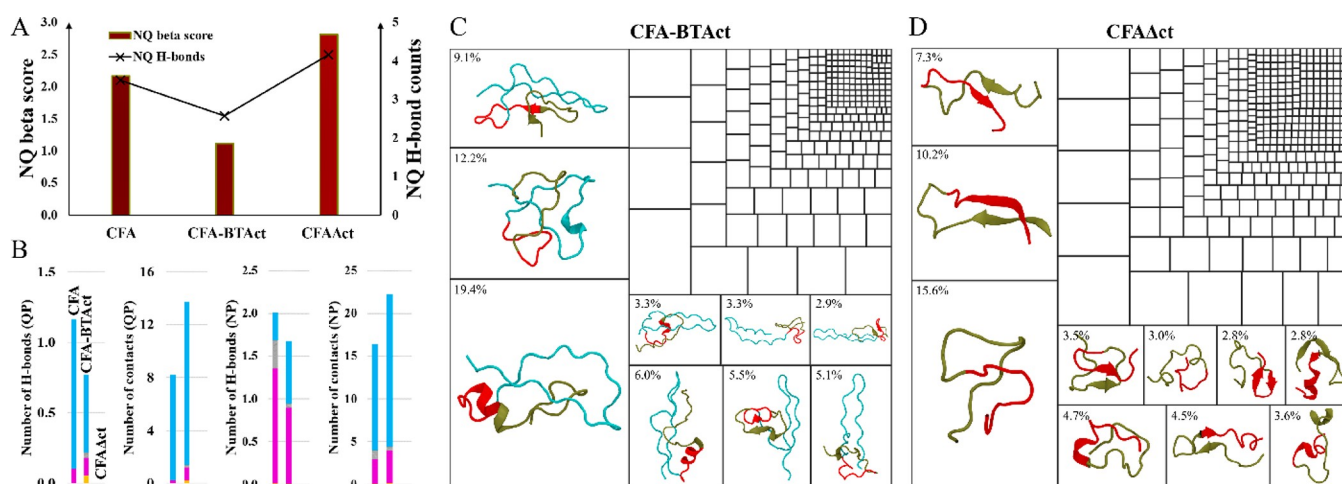


Figure 4. How PRD affects the folding states: a case study of CFA. (A) NQ β scores and NQ H-bond counts (see the definitions in Figure 2) of wild-type exon 1 of CFA Htt (CFA), CFA Htt in which the PRD is replaced with the PRD from BTA Htt (CFA-BTAct), and CFA Htt in which the PRD is truncated (CFA Δ ct). (B) Total number of QP H-bonds, QP contacts, NP H-bonds, and NP contacts for the three peptides (from left to right in each bar plot, CFA, CFA-BTAct, and CFA Δ ct, respectively). (C and D) Tree maps with cluster centers for CFA-BTAct and CFA Δ ct, respectively.

glutamines (Q) from the polyQ domain are heavily involved in the polyQ–PRD interdomain interactions (Figures S7 and S8). This type of nonspecific polyP binding partner with dual interacting properties, that is, H-bond donors and hydrophobic residues (especially aromatic residues),⁶¹ was previously observed in our work on the human profilin–P₁₀ interactions, as well.²⁶ Notably, PQE-1, a proline/glutamine-rich peptide (just like HttEx1-PRD), was previously proposed as a possible treatment for HD due to its ability to weaken the tendency of a HttEx1 mutant to aggregate,^{62,63} albeit no stable PQE-1–HttEx1 binding structures were detected in pull-down assays, an indication that PQE-1–HttEx1 interactions are probably transient. Likewise, when we inspect the residence time of all H-bonds in our simulations, we see that a majority (>90%) of the N17–PRD and polyQ–PRD H-bond pairs persist for <10 ns (Figure S9), indicating the PRD could have the same effects as PQE-1 with regard to influencing the polyQ conformations. Such transient interactions might explain both the flexibility of HttEx1 found by Urbanek et al.⁴² and the long-range N17–PRD contacts found by Warner et al.⁵⁸ Interestingly, the longest consecutive polyP length found in PQE-1 is 4, which raises the question of the ideal polyP length (if any) in PDR to optimally “chaperone” polyQ in the HttEx1.

To search for the “golden ratio” between polyP and polyQ, we start with the outliers in those plots generated from our analyses. At a glance, the polyPs in CFA should have worked, if one assumes that a larger number of prolines would provide better protection (Figure S1D). However, CFA behaves consistently poorly in displaying more interdomain H-bonds and a higher β score in the N17–polyQ case (Figure 1D). Moreover, the CFA systems show higher-polyQ β -content structures (Figure S3) as well as less H-bond formation in the polyQ–PRD and N17–PRD interdomains relative to the respective trend line (Figure 2F). We then notice that in the engineered peptide CFA-BTAct, in which we replaced the PRD of CFA with the PRD of BTA, both the number of H-bonds and the β score in the N17–polyQ interdomain decrease (Figure 4A). This can be attributed to the stronger interactions established by the PRD with the polyQ and N17 domains (therefore, a stronger chaperoning effect from the

PRD) in CFA-BTAct than in native CFA, as shown in Figure 4B. Phenomenologically, the PRD of the peptide CFA-BTAct folds into a proper polyP cage that adequately chaperones the N17 and polyQ domains (Figure 4C). This is in contrast to the more rigid native PRD from the CFA system (Figure 2D). The major difference in the PRD between CFA and BTA lies in the length of the primary polyP (the first polyP stretch): seven-residue segment in the former and five-residue segment in the latter. Meanwhile, the Q length of CFA is 10 residues and that of BTA is 15 residues. From these observations, we speculate that there is an optimal primary polyP length for preventing wild-type HttEx1 from aggregating. If the primary polyP is too long, as in the CFA case, the rigidity of the polyP segment alone may prevent the N17 and polyQ domains from reaching the relevant functional groups, including the glutamines in the PRD (H-bond anchors) and the other polyP stretches (chaperones). On the contrary, if the PRD is nonexistent, the β -strand conformations are more prevalent in peptides as shown in the case of the CFA Δ ct and HQ22 Δ ct systems. These findings from our study provide a logically compelling argument to support a golden ratio of approximately 2:1 for the polyQ domain and the primary polyP segment lengths (as shown in Figure S1D) and the positive correlation between the propensity of prolines at the C-terminus of polyQ and the length of polyQ in the human genome.⁴² Here we propose that a nearly optimal co-evolution for these domains has been reached. This golden ratio allows the PRD to form a proper polyP cage that chaperones the N17 and polyQ domains into forming conformations with fewer β -strands, as further supported by the fact that once the CFA is replaced by CFA-BTAct, those “outliers” are essentially eliminated from the plots of the primary polyP length, number of H-bonds, and β score for the N17–polyQ interdomain (Figure S10).

In tandem with the reduction in β -strand content, the PRD chaperoning capabilities increase the number of helical conformations sampled by the polyQ segments. The α score (similarity of the structure with that of an α -helix⁵⁶) of HQ22 is higher than that of HQ22-P10A, which in turn is higher than that of the HQ22 Δ ct system (Figure S11). Meanwhile, a peptide analogue with a functioning PRD, as in the case of the

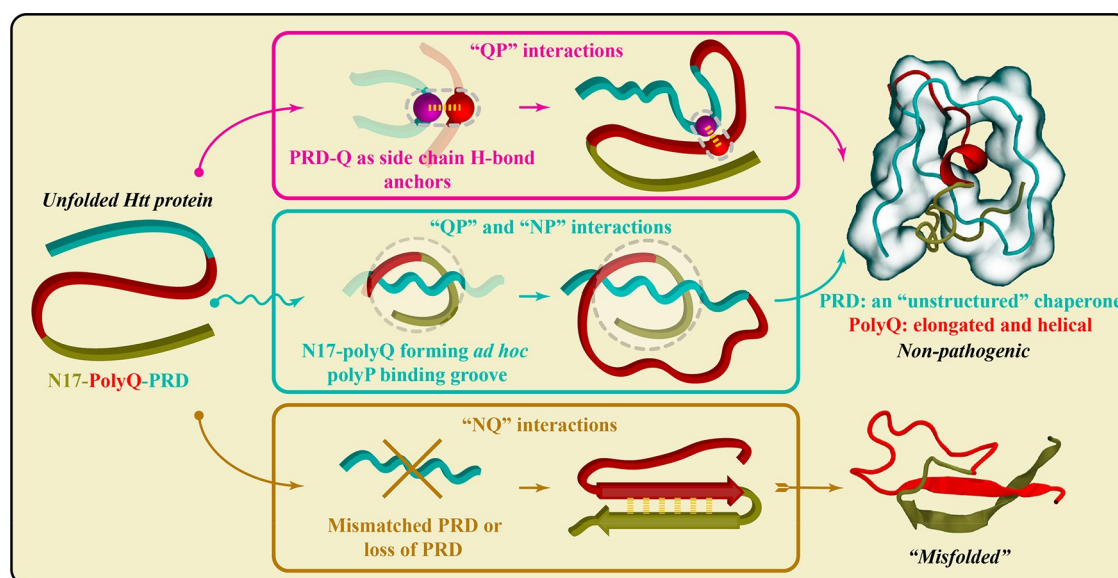


Figure 5. Summarized folding mechanism of Htt exon 1. Tan ribbons represent N17, red ribbons polyQ, and cyan ribbons PRD. PPII helices are highlighted with wavy ribbons, and β -strands are highlighted with thickened arrows. All hydrogen bonds are highlighted with orange dashed lines. Structural cluster 1 (the most populated cluster) of BTA and structural cluster 3 of HQ22 Δ ct are shown as examples of folded proteins.

CFA-BTact system, also displays an α score higher than that of native CFA (Figure S11). Regardless of the increase or decrease in the secondary structure scores, it should be noted that all such secondary structures that require H-bond formation, such as β -sheets, β -bridges, and α -helices, are found to be transient in nature (Figure S9). On the contrary, the major secondary structural components of residues in the N17 and polyQ segments remain as flexible random coiled structure (Figures S12–S14).

Even though there has been progress in understanding the physiopathology of HD, the lack of a complete understanding of the functionality of different domains in the Htt protein has hampered the overall progress of finding an adequate treatment for the disease. In particular, the molecular mechanism by which N17 and the PRD mediate the conformations sampled by the polyQ domain remains largely unknown. In this paper, we proposed two roles of the PRD as an interacting platform in preventing the formation of potentially toxic β -sheets in the Htt protein: (1) a hydrophobic cage capable of allocating nonpolar residues and (2) a hydrophilic locus capable of establishing H-bonds. Specifically, as illustrated in Figure 5, we found that the PRD forms a reasonably rigid and hydrophobic “polyP cage” (with PPII helices) that “chaperones” N17 and polyQ, which results in less β -structure in the HttEx1 monomer. Additionally, a tight relationship between the length of the polyQ domain and the primary polyP stretch might have evolutionary needs as its basis (with the optimal length ratio of polyQ to primary polyP segments of 2:1 resulting in the most functional structure). A more comprehensive understanding of this evolution-driven relationship may provide systematic support for the development of a potential treatment for HD and other protein aggregation-associated diseases.

Our analysis also indicated that apart from the entire PRD, the primary polyP segment alone can function as a structural chaperone, as supported by the results from the HQ22-P10A system and a previously published study.¹⁵ To achieve maximal chaperoning interactions and the subsequent optimal functionality of the protein, the length of the primary polyP should be

commensurate with the length of the polyQ. Specifically, the interactions between the N17 and polyQ segments with the prolines in the PRD occur via the folding of such segments into ad hoc polyP binding grooves, via two types of indiscriminate interactions:^{23–26,28,29} (i) the formation of H-bonds between the backbones of the prolines in the PRD and N17–polyQ and (ii) hydrophobic interactions between the side chain of prolines in the PRD and hydrophobic residues in the N17 domain. The glutamines in the PRD are responsible for establishing strong side chain H-bond interactions and therefore serve as anchors in the PRD. Such a design also reduces the probability of β -structure formation by decreasing the chance of backbone–backbone H-bonds required during the appearance and stabilization of high- β -strand content conformations. It is worth mentioning that the PRD seems to serve as a “negative design” that prevents edge-to-edge aggregation of polyQ, by providing chaperoning effects channeled between less obvious residues (Qs and Ps) compared to previous findings that focused on hydrophobic–hydrophobic and charge–charge interactions.⁵⁰

In summary, in the absence of the PRD, the N17 and polyQ segments are much more likely to adopt β -structures that may lead to aggregation, which induces HD. The PRD therefore plays an essential role in regulating the protein folding of HttEx1 by decreasing the protein’s β -structure content and, in turn, increasing the protein’s α helicity. The reason behind the HttEx1’s need to adopt slightly helical conformations while remaining mostly unstructured, however, remains unclear. One possibility is that such dynamics of conformations may lower the chance of aggregation and subsequent pathology. Nevertheless, our finding suggests that the PRD functions as more than a protein–protein interacting mediator as commonly perceived.⁶⁴ In fact, the PRD is capable of chaperoning surrounding domains while maintaining its PPII helical structure.⁶⁵ Further in-depth investigation on the optimal length ratio between polyP and polyQ in different species would shed light on our understanding of the detailed functionalities of PRD, while examinations of the efficacy of PRD-based treatments and possible side effects hold the

potential of testing the proposed functionality of the PRD in therapeutical use. While the therapeutic possibilities of PRDs in HD^{62,63} (and Alzheimer's disease⁶⁶) were explored previously, we believe that our mechanism-based findings can provide additional and novel insight into the development of more effective curative treatments.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpclett.2c01184>.

Methods, including molecular dynamics simulation, shape anisotropy calculation, and α and β score calculation; additional discussions of PPII helical content in polyQ; an expanded sequence alignment of HttEx1 for 10 species (Figure S1); radii of gyration and relative shape anisotropy for the CFA peptides (Figure S2); average number of residues in β -structures in polyQ domains (Figure S3); NP and QP H-bonds and contacts (Figures S4–S8); interdomain H-bond histogram (Figure S9); polymeric properties of the CFA-BTact peptide (Figure S10); α scores of N17–polyQ (Figure S11); secondary structure propensity of all residues (Figures S12–S14); free energy landscapes of HttEx1 (Figure S15); H-bond and contact frequency maps (Figures S16 and S17); free energy landscapes of polyQ (Figures S18–S23); and sequence alignment showing possible co-slippage in various genes (Table S1) (PDF)

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Author Contributions

R.Z. and L.Z. conceived and designed the study. L.Z. and H.K. performed molecular dynamics simulations. L.Z. collected and analyzed data. L.Z., H.K., J.M.P.-A., and R.Z. interpreted

results and co-wrote the manuscript. All authors contributed to the general discussion of the project and manuscript.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

- (1) Trotter, Y.; Lutz, Y.; Stevanin, G.; Imbert, G.; Devys, D.; et al. Polyglutamine expansion as a pathological epitope in huntington's disease and four dominant cerebellar ataxias. *Nature* **1995**, 378 (6555), 403–406.
- (2) Legleiter, J.; Mitchell, E.; Lotz, G. P.; Sapp, E.; Ng, C.; et al. Mutant huntingtin fragments form oligomers in a polyglutamine length-dependent manner in vitro and in vivo. *J. Biol. Chem.* **2010**, 285 (19), 14777–14790.
- (3) Paul, B. D.; Sbodio, J. I.; Xu, R.; Vandiver, M. S.; Cha, J. Y.; et al. Cystathionine γ -lyase deficiency mediates neurodegeneration in huntington's disease. *Nature* **2014**, 509 (7498), 96–100.
- (4) Cattaneo, E.; Zuccato, C.; Tartari, M. Normal huntingtin function: An alternative approach to huntington's disease. *Nat. Rev. Neurosci.* **2005**, 6 (12), 919–930.
- (5) Saudou, F.; Finkbeiner, S.; Devys, D.; Greenberg, M. E. Huntingtin acts in the nucleus to induce apoptosis but death does not correlate with the formation of intranuclear inclusions. *Cell* **1998**, 95 (1), 55–66.
- (6) Wexler, N. S. Venezuelan kindreds reveal that genetic and environmental factors modulate huntington's disease age of onset. *Proc. Natl. Acad. Sci. U. S. A.* **2004**, 101 (10), 3498–3503.
- (7) Krobisch, S.; Lindquist, S. Aggregation of huntingtin in yeast varies with the length of the polyglutamine expansion and the expression of chaperone proteins. *Proc. Natl. Acad. Sci. U. S. A.* **2000**, 97 (4), 1589–1594.
- (8) Gruber, A.; Hornburg, D.; Antonin, M.; Krahmer, N.; Collado, J.; et al. Molecular and structural architecture of polyQ aggregates in yeast. *Proc. Natl. Acad. Sci. U. S. A.* **2018**, 115 (15), E3446–E3453.
- (9) Kuemmerle, S.; Gutekunst, C. A.; Klein, A. M.; Li, X. J.; Li, S. H.; et al. Huntingtin aggregates may not predict neuronal death in huntington's disease. *Annals of neurology* **1999**, 46 (6), 842–849.
- (10) Simeoni, S.; Mancini, M. A.; Stenoien, D. L.; Marcelli, M.; Weigel, N. L.; et al. Motoneuronal cell death is not correlated with aggregate formation of androgen receptors containing an elongated polyglutamine tract. *Hum. Mol. Genet.* **2000**, 9 (1), 133–144.
- (11) Duennwald, M. L.; Jagadish, S.; Muchowski, P. J.; Lindquist, S. Flanking sequences profoundly alter polyglutamine toxicity in yeast. *Proc. Natl. Acad. Sci. U. S. A.* **2006**, 103 (29), 11045–11050.
- (12) Guo, Q.; Huang, B.; Cheng, J.; Seefelder, M.; Engler, T.; et al. The cryo-electron microscopy structure of huntingtin. *Nature* **2018**, 555 (7694), 117.

- (13) Neueder, A.; Landles, C.; Ghosh, R.; Howland, D.; Myers, R. H.; et al. The pathogenic exon 1 htt protein is produced by incomplete splicing in huntington's disease patients. *Sci. Rep.* **2017**, *7* (1), 1307.
- (14) Baias, M.; Smith, P. E.; Shen, K.; Joachimiak, L. A.; Žerko, S.; et al. Structure and dynamics of the huntingtin exon-1 n-terminus: A solution nmr perspective. *J. Am. Chem. Soc.* **2017**, *139* (3), 1168–1176.
- (15) Thakur, A. K.; Jayaraman, M.; Mishra, R.; Thakur, M.; Chellgren, V. M.; et al. Polyglutamine disruption of the huntingtin exon 1 n terminus triggers a complex aggregation mechanism. *Nature structural & molecular biology* **2009**, *16* (4), 380.
- (16) Michalek, M.; Salnikov, E. S.; Bechinger, B. Structure and topology of the huntingtin 1–17 membrane anchor by a combined solution and solid-state nmr approach. *Biophysical journal* **2013**, *105* (3), 699–710.
- (17) Zhang, L.; Kang, H.; Vázquez, F. X.; Toledo-Sherman, L.; Luan, B.; et al. Molecular mechanism of stabilizing the helical structure of huntingtin n17 in a micellar environment. *J. Phys. Chem. B* **2017**, *121* (18), 4713–4721.
- (18) Maiuri, T.; Woloshansky, T.; Xia, J.; Truant, R. The huntingtin n17 domain is a multifunctional crm1 and ran-dependent nuclear and cilia export signal. *Human molecular genetics* **2013**, *22* (7), 1383–1394.
- (19) Atwal, R. S.; Xia, J.; Pinchev, D.; Taylor, J.; Epan, R. M.; et al. Huntingtin has a membrane association signal that can modulate huntingtin aggregation, nuclear entry and toxicity. *Human molecular genetics* **2007**, *16* (21), 2600–2615.
- (20) Rockabrand, E.; Slepko, N.; Pantalone, A.; Nukala, V. N.; Kazantsev, A.; et al. The first 17 amino acids of huntingtin modulate its sub-cellular localization, aggregation and effects on calcium homeostasis. *Human molecular genetics* **2007**, *16* (1), 61–77.
- (21) Lin, H.-K.; Boatz, J. C.; Krabbendam, I. E.; Kodali, R.; Hou, Z.; et al. Fibril polymorphism affects immobilized non-amyloid flanking domains of huntingtin exon1 rather than its polyglutamine core. *Nat. Commun.* **2017**, *8*, 15462.
- (22) Hoop, C. L.; Lin, H.-K.; Kar, K.; Magyarfalvi, G.; Lamley, J. M.; et al. Huntingtin exon 1 fibrils feature an interdigitated β -hairpin-based polyglutamine core. *Proc. Natl. Acad. Sci. U. S. A.* **2016**, *113* (6), 1546–1551.
- (23) Southwell, A. L.; Khoshnan, A.; Dunn, D. E.; Bugg, C. W.; Lo, D. C.; et al. Intrabodies binding the proline-rich domains of mutant huntingtin increase its turnover and reduce neurotoxicity. *J. Neurosci.* **2008**, *28* (36), 9013–9020.
- (24) Faber, P. W.; Barnes, G. T.; Srinidhi, J.; Chen, J.; Gusella, J. F.; et al. Huntingtin interacts with a family of ww domain proteins. *Human molecular genetics* **1998**, *7* (9), 1463–1474.
- (25) Liu, Y. F.; Deth, R. C.; Devys, D. Sh3 domain-dependent association of huntingtin with epidermal growth factor receptor signaling complexes. *J. Biol. Chem.* **1997**, *272* (13), 8121–8124.
- (26) Zhang, L.; Bell, D. R.; Luan, B.; Zhou, R. Exploring the binding mechanism between human profilin (pfn1) and polyproline-10 through binding mode screening. *J. Chem. Phys.* **2019**, *150* (1), 015102.
- (27) Lim, W. A.; Richards, F. M.; Fox, R. O. Structural determinants of peptide-binding orientation and of sequence specificity in sh3 domains. *Nature* **1994**, *372* (6504), 375–379.
- (28) Adzhubei, A. A.; Sternberg, M. J.; Makarov, A. A. Polyproline-ii helix in proteins: Structure and function. *Journal of molecular biology* **2013**, *425* (12), 2100–2132.
- (29) Morgan, A. A.; Rubenstein, E. Proline: The distribution, frequency, positioning, and common functional roles of proline and polyproline sequences in the human proteome. *PLoS one* **2013**, *8* (1), e53785.
- (30) Crick, S. L.; Ruff, K. M.; Garai, K.; Frieden, C.; Pappu, R. V. Unmasking the roles of n- and c-terminal flanking sequences from exon 1 of huntingtin as modulators of polyglutamine aggregation. *Proc. Natl. Acad. Sci. U. S. A.* **2013**, *110* (50), 20075–20080.
- (31) Bhattacharyya, A.; Thakur, A. K.; Chellgren, V. M.; Thiagarajan, G.; Williams, A. D.; et al. Oligoproline effects on polyglutamine conformation and aggregation. *Journal of molecular biology* **2006**, *355* (3), 524–535.
- (32) Chellgren, B. W.; Miller, A.-F.; Creamer, T. P. Evidence for polyproline ii helical structure in short polyglutamine tracts. *Journal of molecular biology* **2006**, *361* (2), 362–371.
- (33) Darnell, G.; Orgel, J. P.; Pahl, R.; Meredith, S. C. Flanking polyproline sequences inhibit β -sheet structure in polyglutamine segments by inducing ppII-like helix structure. *Journal of molecular biology* **2007**, *374* (3), 688–704.
- (34) Darnell, G. D.; Derryberry, J.; Kurutz, J. W.; Meredith, S. C. Mechanism of cis-inhibition of polyq fibrillation by polyp: PpII oligomers and the hydrophobic effect. *Biophysical journal* **2009**, *97* (8), 2295–2305.
- (35) Duennwald, M. L.; Jagadish, S.; Giorgini, F.; Muchowski, P. J.; Lindquist, S. A network of protein interactions determines polyglutamine toxicity. *Proc. Natl. Acad. Sci. U. S. A.* **2006**, *103* (29), 11051–11056.
- (36) Morovvati, S.; Nakagawa, M.; Osame, M.; Karami, A. Analysis of cag repeats in huntingtin gene among hd patients and normal populations in japan. *Archives of medical research* **2008**, *39* (1), 131–133.
- (37) Lakhani, V. V.; Ding, F.; Dokholyan, N. V. Polyglutamine induced misfolding of huntingtin exon1 is modulated by the flanking sequences. *PLoS Comput. Biol.* **2010**, *6* (4), e1000772.
- (38) Chen, M.; Wolynes, P. G. Aggregation landscapes of huntingtin exon 1 protein fragments and the critical repeat length for the onset of huntington's disease. *Proc. Natl. Acad. Sci. U. S. A.* **2017**, *114* (17), 4406–4411.
- (39) Moldoveanu, S. N.; Chiş, V. Molecular dynamics simulations applied to structural and dynamical transitions of the huntingtin protein: A review. *ACS chemical neuroscience* **2020**, *11* (2), 105–120.
- (40) Tartari, M.; Gissi, C.; Lo Sardo, V.; Zuccato, C.; Picardi, E.; et al. Phylogenetic comparison of huntingtin homologues reveals the appearance of a primitive polyq in sea urchin. *Molecular biology and evolution* **2008**, *25* (2), 330–338.
- (41) Marcos, M. L.; Echave, J. Too packed to change: Side-chain packing and site-specific substitution rates in protein evolution. *PeerJ.* **2015**, *3*, e911.
- (42) Urbanek, A.; Popovic, M.; Morató, A.; Estaña, A.; Elena-Real, C. A.; et al. Flanking regions determine the structure of the polyglutamine in huntingtin through mechanisms common among glutamine-rich human proteins. *Structure* **2020**, *28* (7), 733–746.e5.
- (43) Seefelder, M.; Alva, V.; Huang, B.; Engler, T.; Baumeister, W.; et al. The evolution of the huntingtin-associated protein 40 (hap40) in conjunction with huntingtin. *BMC Evol. Biol.* **2020**, *20* (1), 162.
- (44) Crick, S. L.; Jayaraman, M.; Frieden, C.; Wetzel, R.; Pappu, R. V. Fluorescence correlation spectroscopy shows that monomeric polyglutamine molecules form collapsed structures in aqueous solutions. *Proc. Natl. Acad. Sci. U. S. A.* **2006**, *103* (45), 16764–16769.
- (45) Vitalis, A.; Wang, X.; Pappu, R. V. Atomistic simulations of the effects of polyglutamine chain length and solvent quality on conformational equilibria and spontaneous homodimerization. *Journal of molecular biology* **2008**, *384* (1), 279–297.
- (46) Kang, H.; Vázquez, F. X.; Zhang, L.; Das, P.; Toledo-Sherman, L.; et al. Emerging β -sheet rich conformations in supercompact huntingtin exon-1 mutant structures. *J. Am. Chem. Soc.* **2017**, *139* (26), 8820–8827.
- (47) Wagner, A. S.; Politi, A. Z.; Ast, A.; Bravo-Rodriguez, K.; Baum, K.; et al. Self-assembly of mutant huntingtin exon-1 fragments into large complex fibrillar structures involves nucleated branching. *Journal of molecular biology* **2018**, *430* (12), 1725–1744.
- (48) Kar, K.; Hoop, C. L.; Drombosky, K. W.; Baker, M. A.; Kodali, R.; et al. B-hairpin-mediated nucleation of polyglutamine amyloid formation. *Journal of molecular biology* **2013**, *425* (7), 1183–1197.
- (49) Nagai, Y.; Inui, T.; Popiel, H. A.; Fujikake, N.; Hasegawa, K.; et al. A toxic monomeric conformer of the polyglutamine protein. *Nature structural & molecular biology* **2007**, *14* (4), 332–340.

(50) Thirumalai, D.; Klimov, D.; Dima, R. Emerging ideas on the molecular basis of protein and peptide aggregation. *Curr. Opin. Struct. Biol.* **2003**, *13* (2), 146–159.

(51) Chakraborty, D.; Straub, J. E.; Thirumalai, D. Differences in the free energies between the excited states of a β 40 and a β 42 monomers encode their aggregation propensities. *Proc. Natl. Acad. Sci. U. S. A.* **2020**, *117* (33), 19926–19937.

(52) Pandey, N. K.; Isas, J. M.; Rawat, A.; Lee, R. V.; Langen, J.; et al. The 17-residue-long n terminus in huntingtin controls stepwise aggregation in solution and on membranes via different mechanisms. *J. Biol. Chem.* **2018**, *293* (7), 2597–2605.

(53) De Genst, E.; Chirgadze, D. Y.; Klein, F. A.; Butler, D. C.; Matak-Vinković, D.; et al. Structure of a single-chain fv bound to the 17 n-terminal residues of huntingtin provides insights into pathogenic amyloid formation and suppression. *Journal of molecular biology* **2015**, *427* (12), 2166–2178.

(54) Newcombe, E. A.; Ruff, K. M.; Sethi, A.; Ormsby, A. R.; Ramdhan, Y. M.; et al. Tadpole-like conformations of huntingtin exon 1 are characterized by conformational heterogeneity that persists regardless of polyglutamine length. *Journal of molecular biology* **2018**, *430* (10), 1442–1458.

(55) Sivanandam, V.; Jayaraman, M.; Hoop, C. L.; Kodali, R.; Wetzel, R.; et al. The aggregation-enhancing huntingtin n-terminus is helical in amyloid fibrils. *J. Am. Chem. Soc.* **2011**, *133* (12), 4558–4566.

(56) Pietrucci, F.; Laio, A. A collective variable for the efficient exploration of protein beta-sheet structures: Application to sh3 and gb1. *J. Chem. Theory Comput.* **2009**, *5* (9), 2197–2201.

(57) Priya, S. B.; Gromiha, M. M. Structural insights into the aggregation mechanism of huntingtin exon 1 protein fragment with different polyq-lengths. *Journal of cellular biochemistry* **2019**, *120* (6), 10519–10529.

(58) Warner, J. B., IV; Ruff, K. M.; Tan, P. S.; Lemke, E. A.; Pappu, R. V.; et al. Monomeric huntingtin exon 1 has similar overall structural features for wild-type and pathological polyglutamine lengths. *J. Am. Chem. Soc.* **2017**, *139* (41), 14456–14469.

(59) Smith, C. K.; Withka, J. M.; Regan, L. A thermodynamic scale for the beta-sheet forming tendencies of the amino acids. *Biochemistry* **1994**, *33* (18), 5510–5517.

(60) Costantini, S.; Colonna, G.; Facchiano, A. M. Amino acid propensities for secondary structures are influenced by the protein structural class. *Biochemical and biophysical research communications* **2006**, *342* (2), 441–451.

(61) Williamson, M. P. The structure and function of proline-rich regions in proteins. *Biochem. J.* **1994**, *297* (2), 249.

(62) Faber, P. W.; Voisine, C.; King, D. C.; Bates, E. A.; Hart, A. C. Glutamine/proline-rich pqr-1 proteins protect *Caenorhabditis elegans* neurons from huntingtin polyglutamine neurotoxicity. *Proc. Natl. Acad. Sci. U. S. A.* **2002**, *99* (26), 17131–17136.

(63) Voisine, C.; Varma, H.; Walker, N.; Bates, E. A.; Stockwell, B. R.; et al. Identification of potential therapeutic drugs for huntington's disease using *Caenorhabditis elegans*. *PLoS one* **2007**, *2* (6), e504.

(64) Zarrinpar, A.; Bhattacharyya, R. P.; Lim, W. A. The structure and function of proline recognition domains. *Sci. Signaling* **2003**, *2003* (179), re8–re8.

(65) Reiersen, H.; Rees, A. R. The hunchback and its neighbours: Proline as an environmental modulator. *Trends in biochemical sciences* **2001**, *26* (11), 679–684.

(66) Janusz, M.; Zablocka, A. Colostral proline-rich polypeptides-immunoregulatory properties and prospects of therapeutic use in Alzheimer's disease. *Current Alzheimer Research* **2010**, *7* (4), 323–333.

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